

Advanced World Model Directions

for Decision Making

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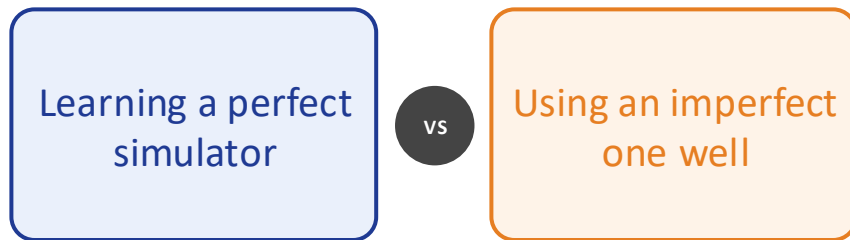
May 13, 2026

Two questions before we begin

Q1 · World models — for what?

- a Sample-efficient policy learning
- b Long-horizon planning
- c Safe / verifiable behaviour
- d Simulating new data

Q2 · Which is harder?



Roadmap



Dreamer and its legacy

DreamerV3, Director,
Disagreement,
SafeDreamer



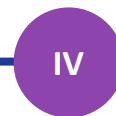
Explicit vs implicit

TD-MPC2



Video models & VLAs

Cosmos Policy, 1X NEO

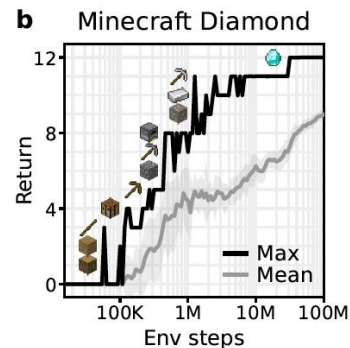
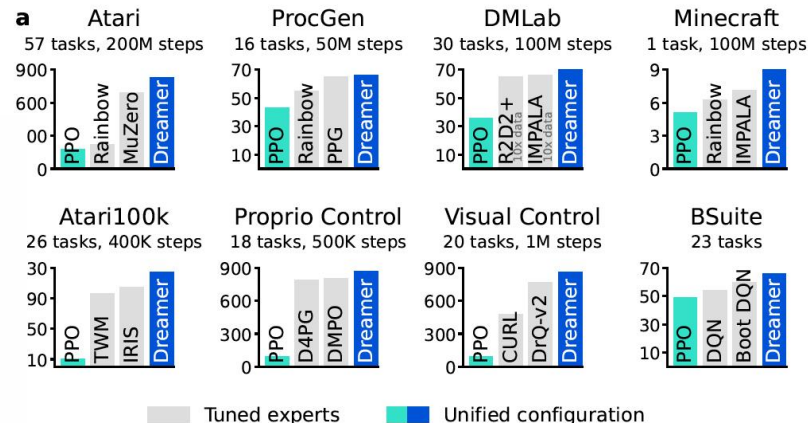
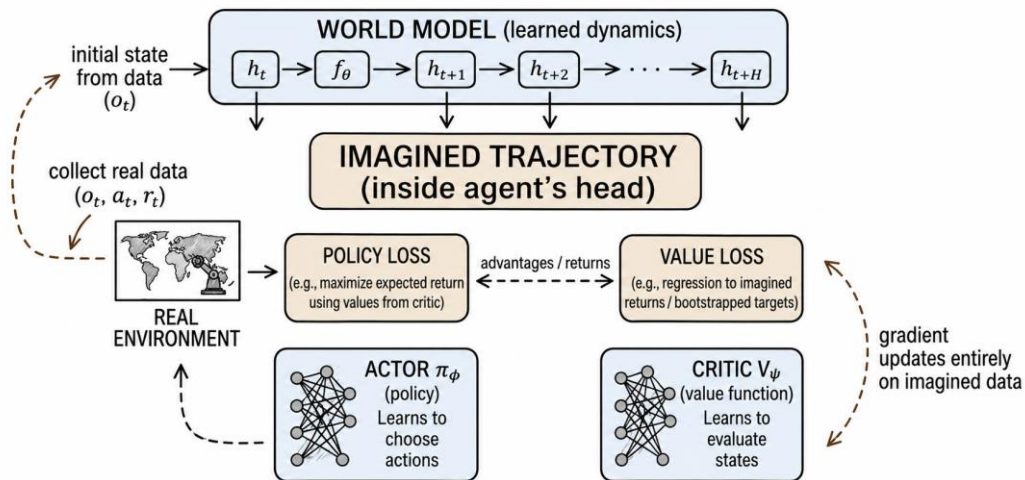


World model as simulator

DreamDojo



Dreamer · three networks, one imagination loop



Recurrent State-Space Model (RSSM)

World model loss

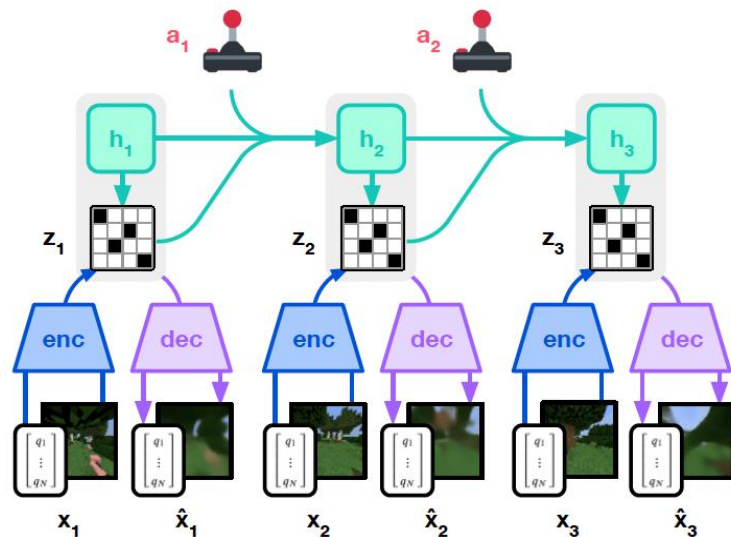
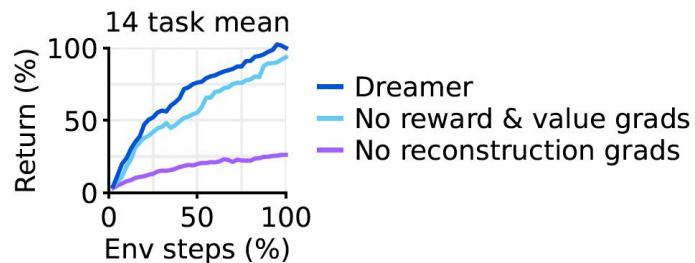
$$\mathcal{L}(\phi) \doteq \mathbb{E}_{q_\phi} \left[\sum_{t=1}^T (\beta_{\text{pred}} \mathcal{L}_{\text{pred}}(\phi) + \beta_{\text{dyn}} \mathcal{L}_{\text{dyn}}(\phi) + \beta_{\text{rep}} \mathcal{L}_{\text{rep}}(\phi)) \right]$$

$$\mathcal{L}_{\text{pred}}(\phi) \doteq -\ln p_\phi(x_t | z_t, h_t) - \ln p_\phi(r_t | z_t, h_t) - \ln p_\phi(c_t | z_t, h_t)$$

$$\mathcal{L}_{\text{dyn}}(\phi) \doteq \max(1, \text{KL}[\text{sg}(q_\phi(z_t | h_t, x_t)) \parallel p_\phi(z_t | h_t)])$$

$$\mathcal{L}_{\text{rep}}(\phi) \doteq \max(1, \text{KL}[q_\phi(z_t | h_t, x_t) \parallel \text{sg}(p_\phi(z_t | h_t))])$$

b Learning signals



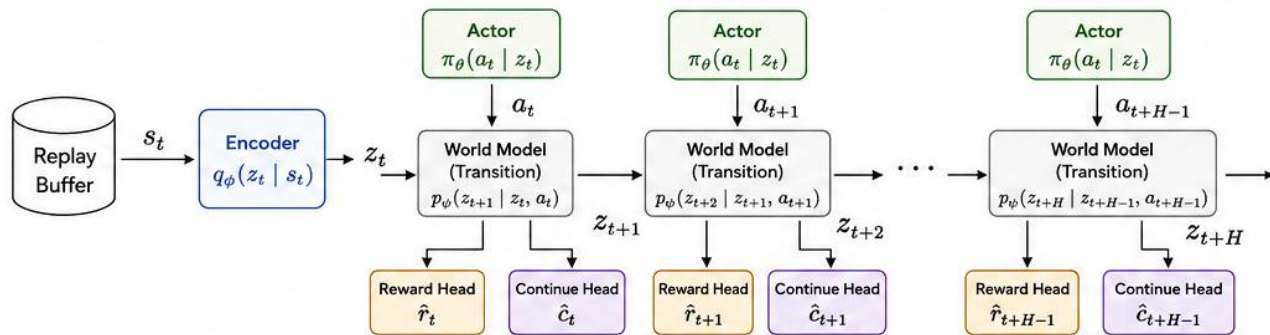
Learning in imagination, not in the environment

1. Start from a real state

Sample s_t from replay buffer and encode.

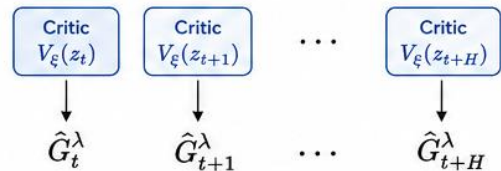
2. Imagine forward for H steps using the actor

Roll out the world model to get imagined trajectory of latent states, predicted rewards and continue-flags.



3. Compute values and λ -returns

Compute critic value at every step and combine rewards with bootstrapped values into λ -returns.



4. Update actor

Backpropagate through the dynamics—straight-through gradients through the latent.

$$\nabla_{\theta} J_{\text{actor}}$$

5. Update critic

Regression to λ -returns.

$$\nabla_{\xi} \mathcal{L}_{\text{critic}}$$

λ -Returns (with continue): Let $\gamma \in (0, 1)$, $\lambda \in [0, 1]$, and $\hat{c}_k \in [0, 1]$ be the predicted continue (discount) probability.

$$\hat{G}_T^{\lambda} = V_{\xi}(z_T) \quad \hat{G}_t^{\lambda} = \hat{r}_t + \gamma \hat{c}_t [(1 - \lambda)V_{\xi}(z_{t+1}) + \lambda \hat{G}_{t+1}^{\lambda}], \quad t = T - 1, \dots, t_0$$

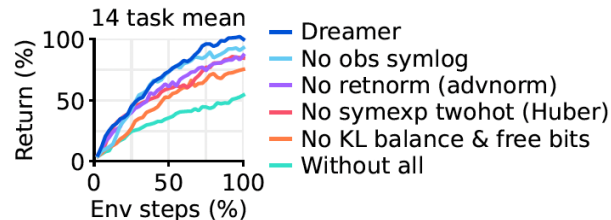
$\hat{r}_t$: predicted reward
 $\hat{c}_t$: predicted continue (discount)
 $V_{\xi}(z_t)$: critic value

$$\mathcal{L}(\theta) \doteq - \sum_{t=1}^T \text{sg} \left(\hat{G}_t^{\lambda} - v_{\psi}(s_t) \right) / \max(1, S) \log \pi_{\theta}(a_t | s_t) + \eta H [\pi_{\theta}(a_t | s_t)]$$

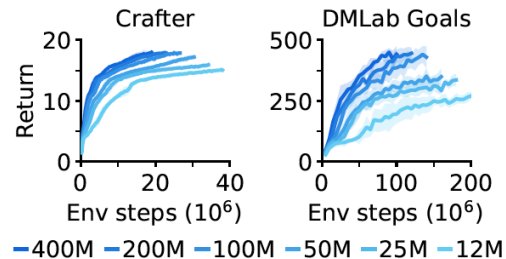
What makes it scale · fixed hyperparameters, predictable returns

- ✓ Symlog observation / reward transforms
- ✓ Two-hot regression for reward and value
- ✓ KL balancing + free bits
- ✓ Percentile return normalisation
- ✓ Adaptive gradient clipping
- ✓ Block-diagonal GRU

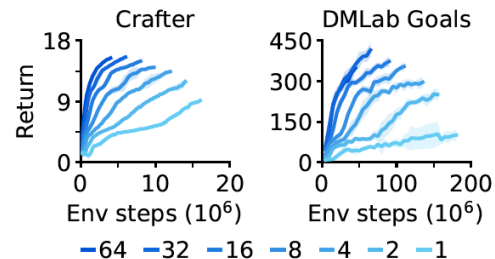
a Robustness techniques



c Model size scaling



d Replay scaling



Dreamer · what it wins and what it pays

Pros

- + General — same hyperparameters across domains
- + Sample-efficient — actor-critic in imagination
- + Interpretable — decode imagined latents to images
- + Predictable scaling with model size & replay ratio

Cons

- Reconstruction is expensive · capacity spent on irrelevant pixels
- Imagination horizon limited (~15 steps) · long horizons need help

→ Part 2 questions reconstruction. → Next slide solves the horizon problem.

The long-horizon problem

10,000+

primitive actions to make a cup of tea

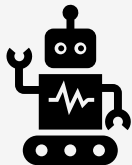
Dreamer ≈ 15

100,000+

primitive actions to collect a diamond in Minecraft



Flat RL Agent



reasons at the rate of primitive actions

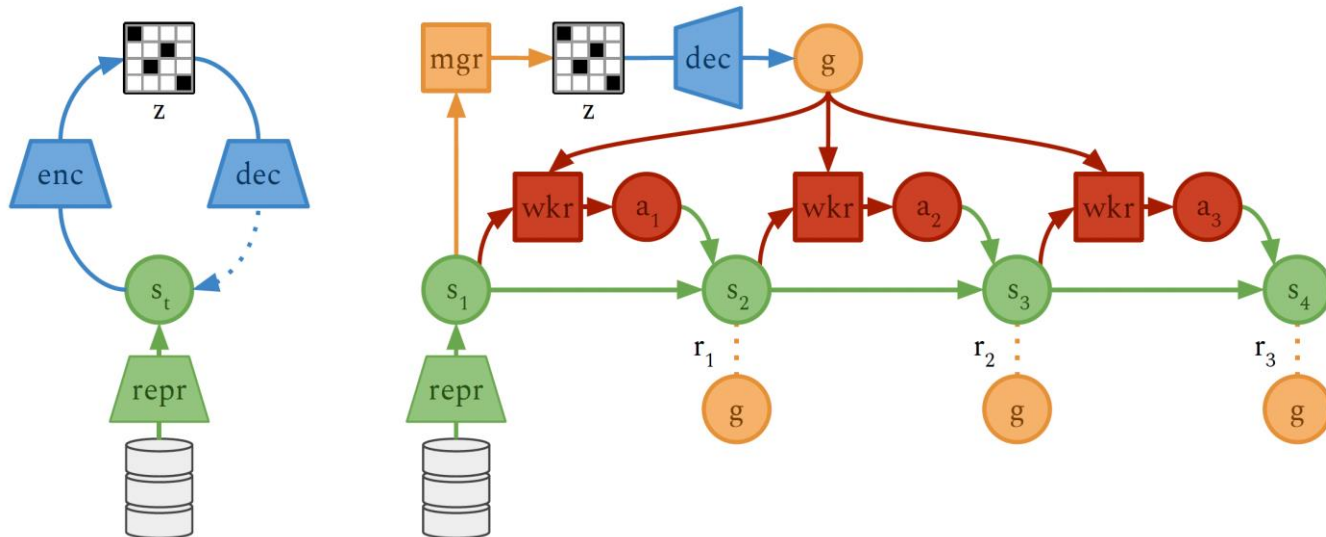
Human



→ kitchen → cupboard → teapot
reasons in subgoals

Can we get an agent to learn the subgoals automatically?

Director · hierarchy on top of a Dreamer world model



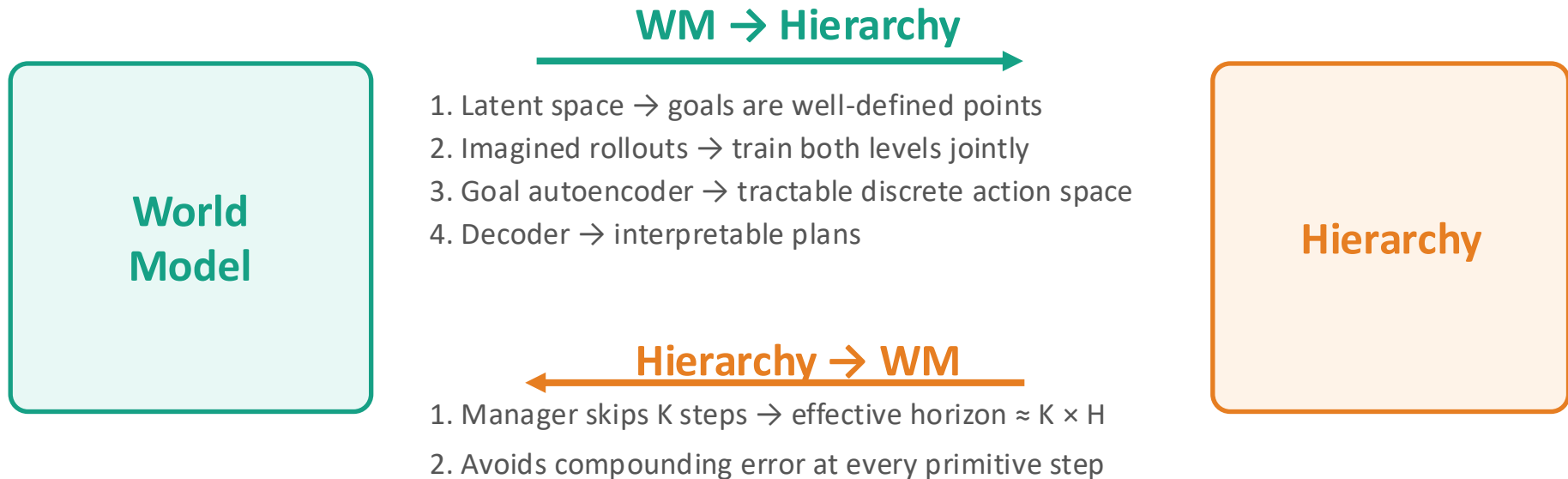
Worker
goal conditioned
actor policy

Manager
actor policy
define goals

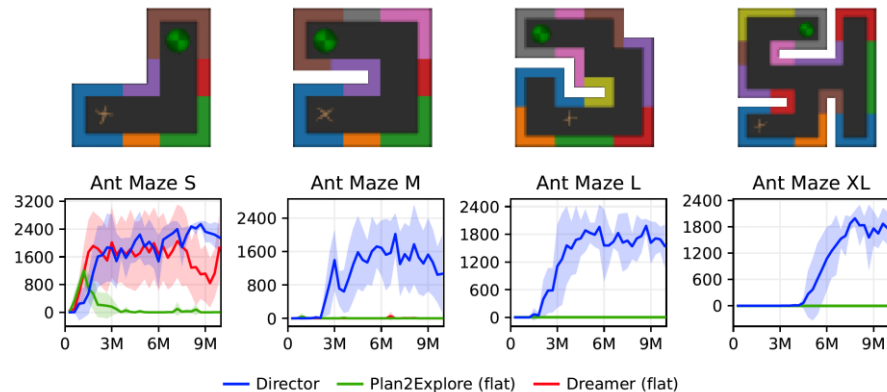
Goal autoencoder

World model: same as Dreamer's RSSM.

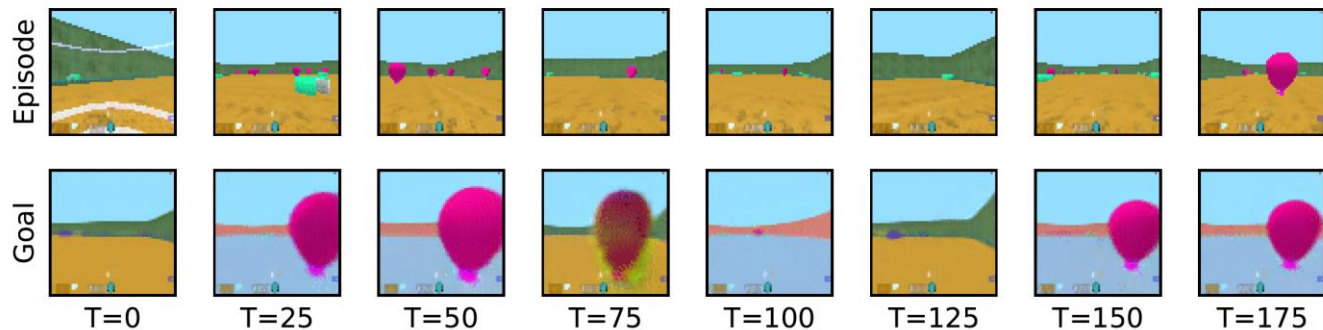
Why hierarchy and the world model need each other



Director · sparse-reward wins + interpretable subgoals

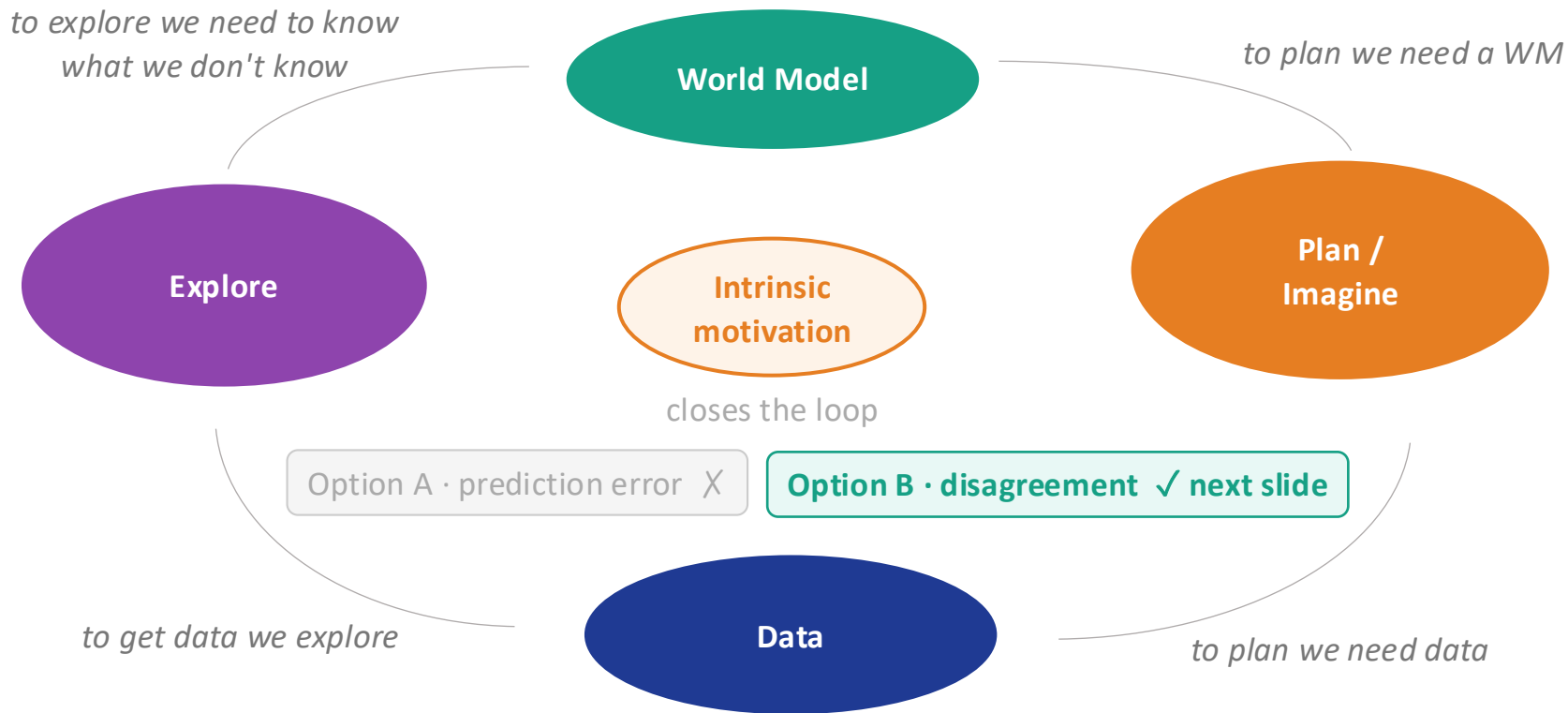


Decoded manager goals over an episode

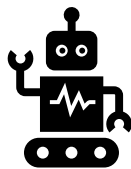
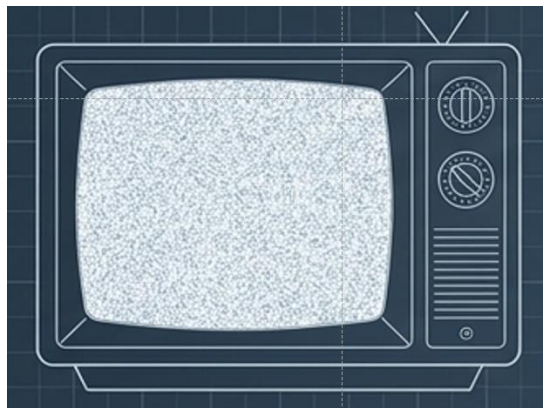


Decoded latent goals are semantically meaningful: inventory items, body poses, locations.

Exploration and the world model · a closed loop



Why prediction error fails · the noisy TV



Wow, every
channel is new!

All 'noisy TVs' in the real world:

keychain (stochasticity)

shadows of leaves

glass reflections

Two kinds of uncertainty

Aleatoric
in the environment
irreducible

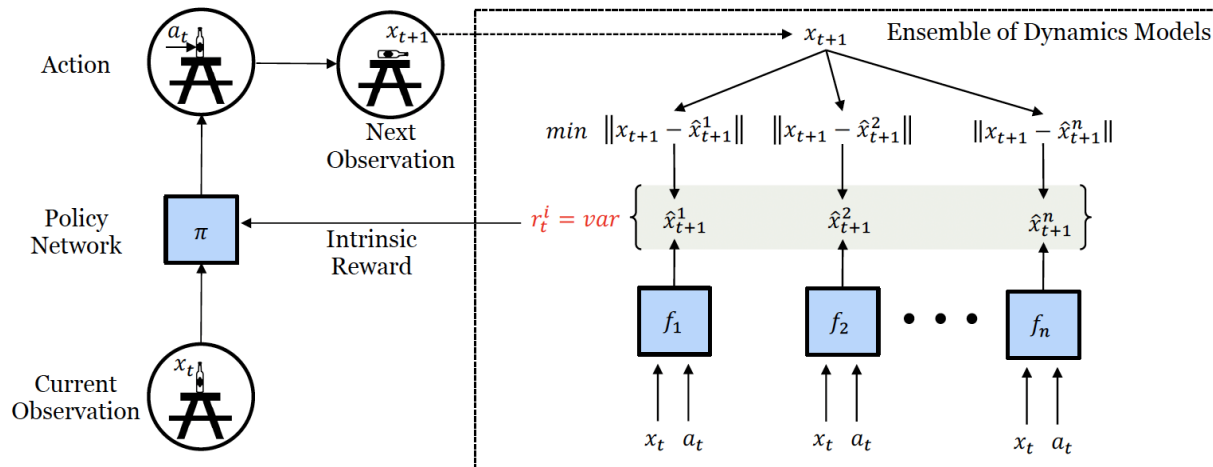
Epistemic
in the model
reducible by data

Prediction error confounds them

prediction error = aleatoric + epistemic

☹ agent rewards itself for stochasticity

Disagreement · the variance of an ensemble



On the noisy TV

All three models predict the same.
Variance $\rightarrow 0$. Reward $\rightarrow 0$.

On genuine novelty

Models disagree — predictions diverge.
Variance high. Reward high.

Bonus: differentiable through the model. Direct policy gradient, not REINFORCE.

Disagreement plugs into Dreamer

Dreamer World Model

Disagreement

Plan2Explore

Director



Atari (sticky)

Outperforms prediction-error on Pong, Breakout, Enduro



Real robot

67% interaction with unseen objects vs 17% random vs 1% REINFORCE



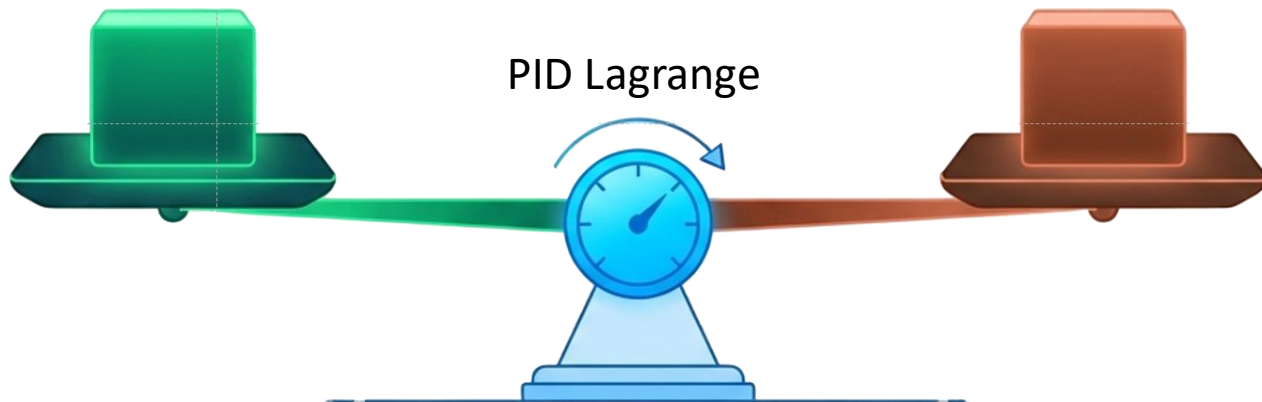
Director context

Disagreement reward at the manager → temporally-abstract exploration

Safety in RL · the Constrained MDP

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right]$$

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right]$$



Why hard?

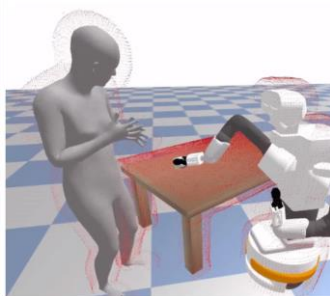
Lagrangian methods oscillate. Vision input has no explicit dynamics.

Why the world model helps

Next slide →

Dangerous imagination · safe execution

In the world model



Cost of trying = GPU seconds

On the real robot

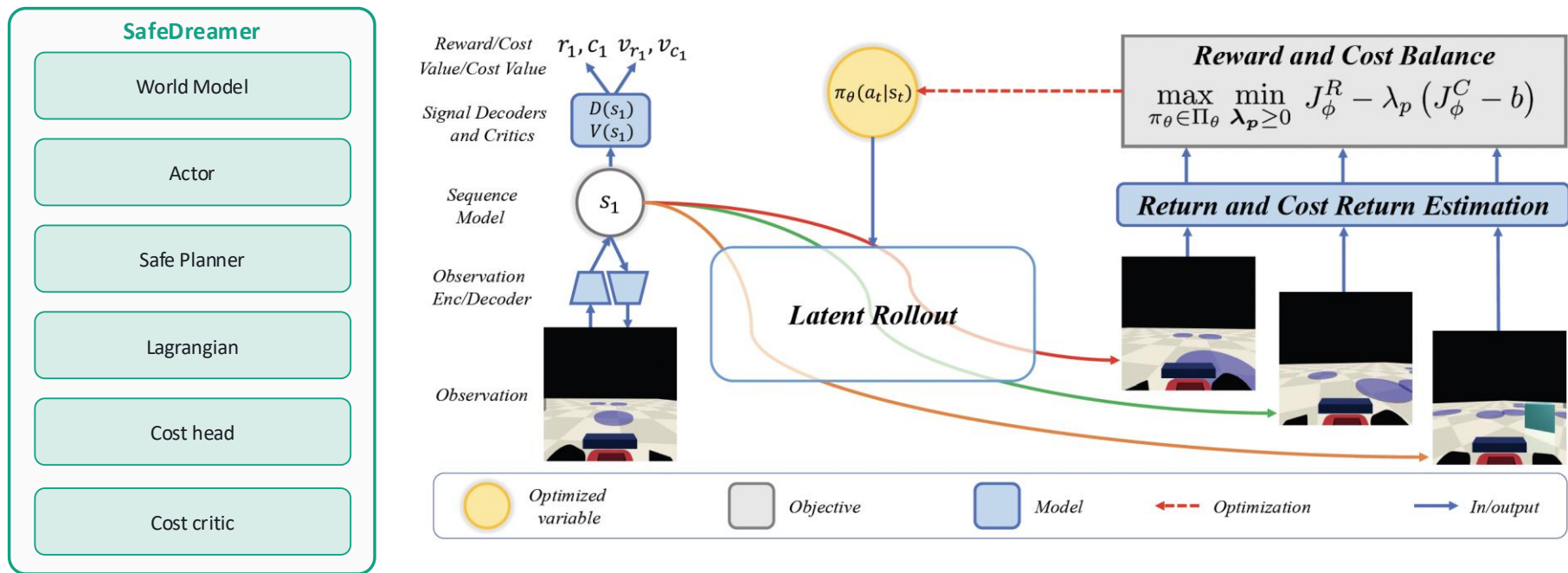


navigating safely
past obstacles ✓

Cost of trying = real damage, real harm

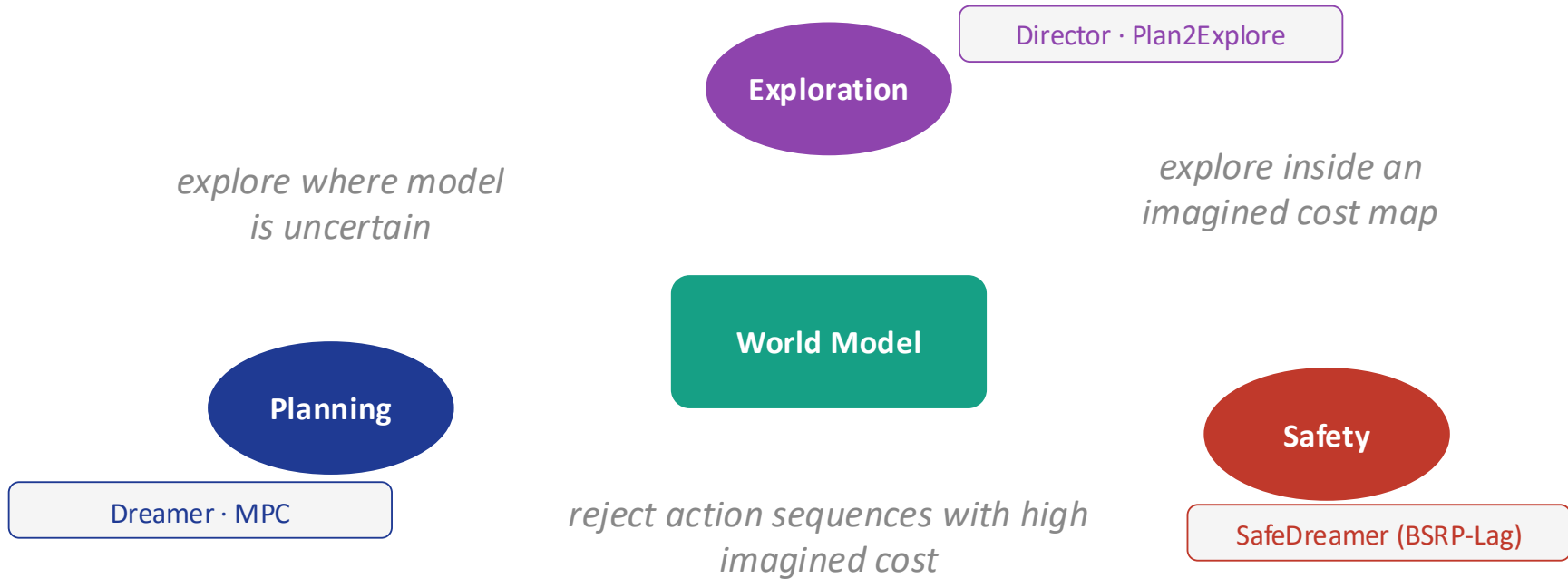
Architecture delta: + cost head + cost critic + safe planner

SafeDreamer · three flavours of safety-reward planning



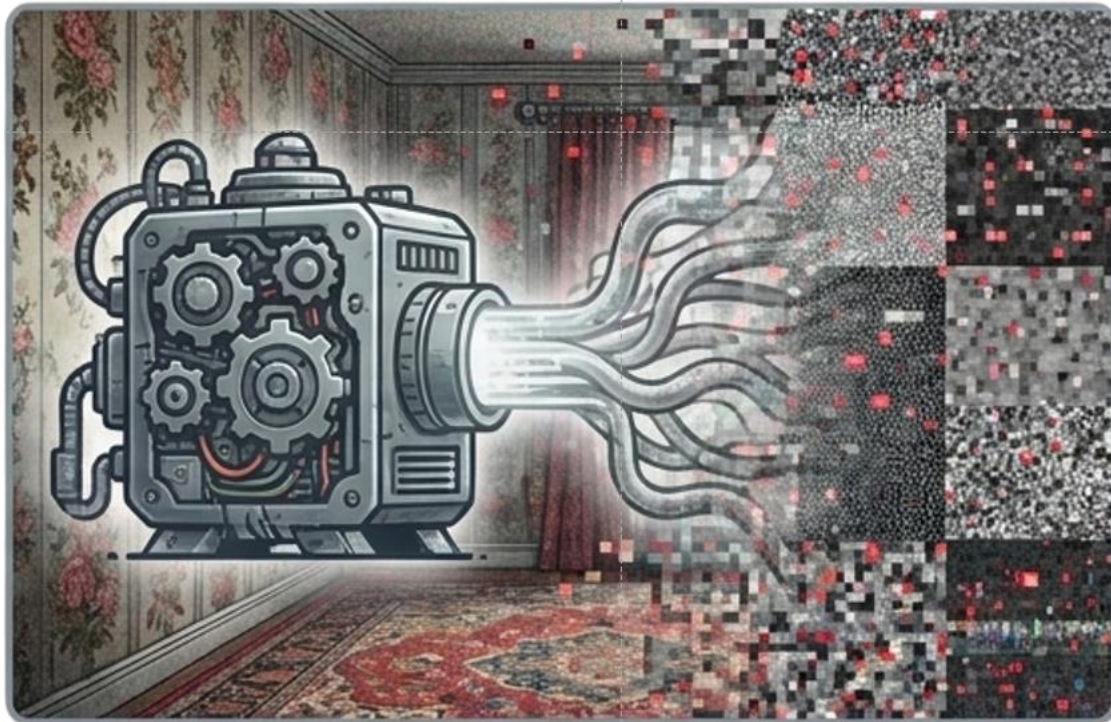
Near-zero cost on Safety-Gymnasium · first vision-based algorithm to achieve this.

The triangle · three problems, one architectural locus



→ Part 2 · what if we throw away the decoder?

The objective-mismatch problem



World Model
Learning



Task Learning

Do we need to decode at all?

Two world-model branches

World Model

Explicit · with decoder

Pros

Rich self-supervised signal · Decodable · Works from scratch

Cons

Decoder cost · Wasted capacity · Objective mismatch

Dreamer

Director

SafeDreamer

Cosmos

Implicit · no decoder

Pros

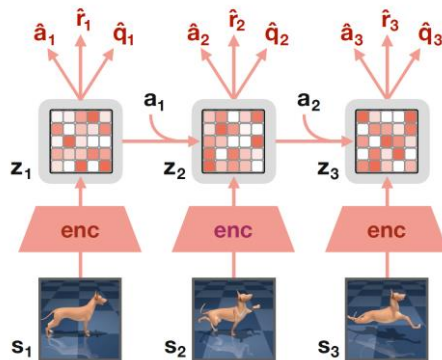
Control-centric · Cheap · Scales to multitask

Cons

Harder to debug · Lose reconstruction signal

TD-MPC2

TD-MPC2 · five components, no decoder

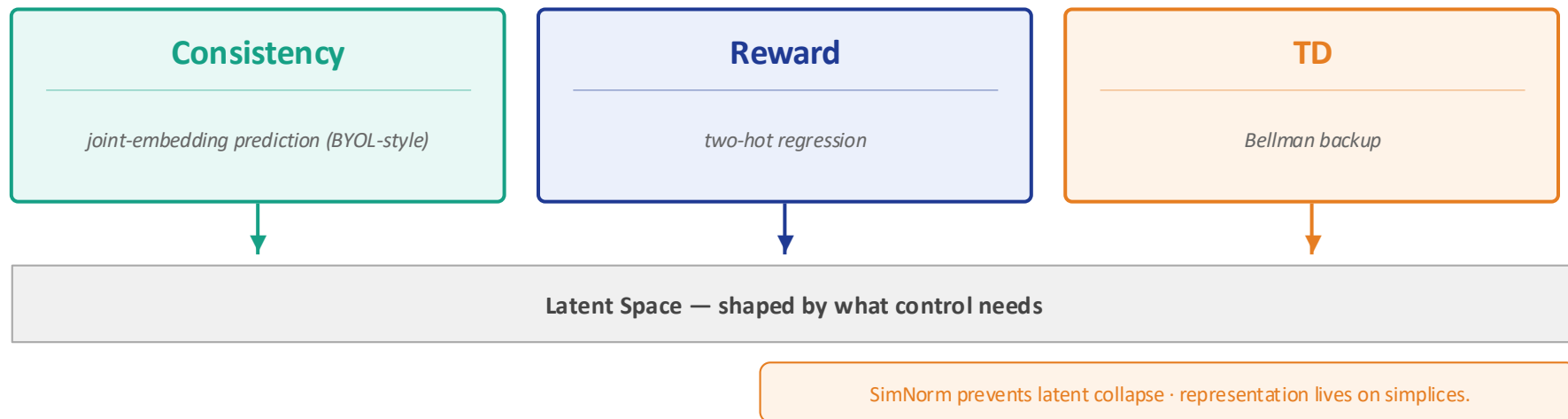


Encoder	$\mathbf{z} = h(\mathbf{s}, \mathbf{e})$	▷ Maps observations to their latent representations
Latent dynamics	$\mathbf{z}' = d(\mathbf{z}, \mathbf{a}, \mathbf{e})$	▷ Models (latent) forward dynamics
Reward	$\hat{r} = R(\mathbf{z}, \mathbf{a}, \mathbf{e})$	▷ Predicts reward r of a transition
Terminal value	$\hat{q} = Q(\mathbf{z}, \mathbf{a}, \mathbf{e})$	▷ Predicts discounted sum of rewards (return)
Policy prior	$\hat{\mathbf{a}} = p(\mathbf{z}, \mathbf{e})$	▷ Predicts action $\mathbf{a}^*$ that maximizes Q

No decoder anywhere in this list.

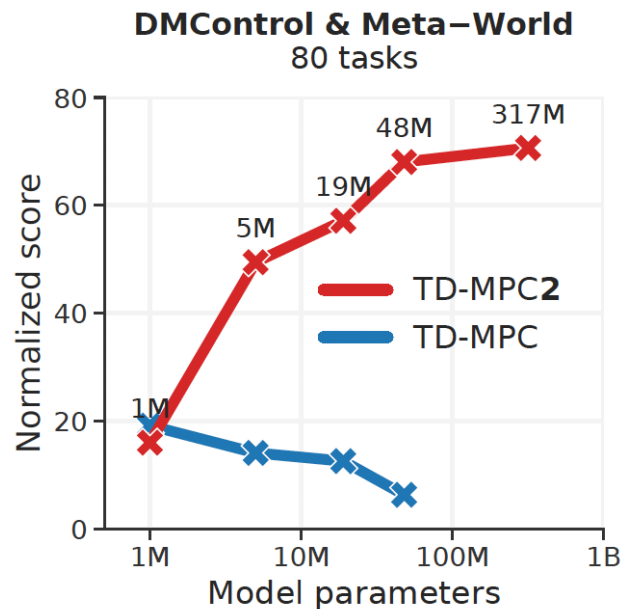
Training the implicit model · three losses, no reconstruction

$$\mathcal{L}(\theta) \doteq \mathbb{E}_{(\mathbf{s}, \mathbf{a}, r, \mathbf{s}')_{0:H} \sim \mathcal{B}} \left[\sum_{t=0}^H \lambda^t \left(\underbrace{\|\mathbf{z}'_t - \text{sg}(h(\mathbf{s}'_t))\|_2^2}_{\text{Joint-embedding prediction}} + \underbrace{\text{CE}(\hat{r}_t, r_t)}_{\text{Reward prediction}} + \underbrace{\text{CE}(\hat{q}_t, q_t)}_{\text{Value prediction}} \right) \right]$$

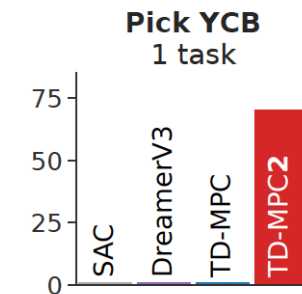
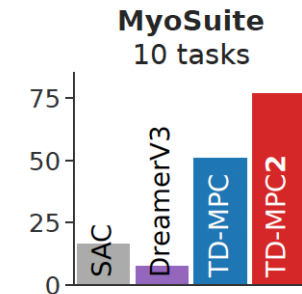
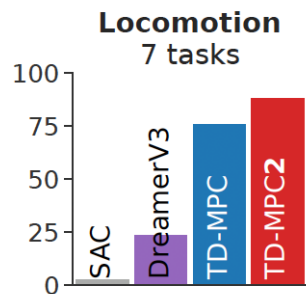
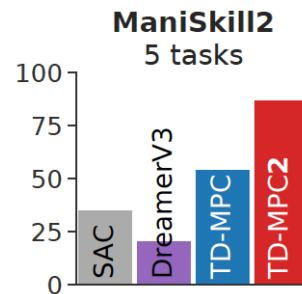
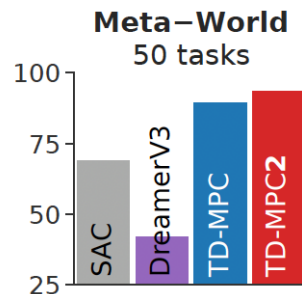
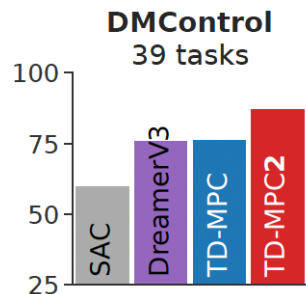


TD-MPC2 · one set of hyperparameters, 104 tasks

Multi-task



Single-task



When does each branch win?

Pick explicit when...

- ✓ Data scarce — reconstruction is free auxiliary signal
- ✓ You need interpretability — see what the agent imagines
- ✓ Bootstrap from scratch on hard sparse-reward domains

Example: Dreamer on Minecraft diamond

Pick implicit when...

- ✓ Data abundant — multitask, many trajectories
- ✓ Observation space has lots of irrelevant detail
- ✓ Fine control where pixel precision is wasted

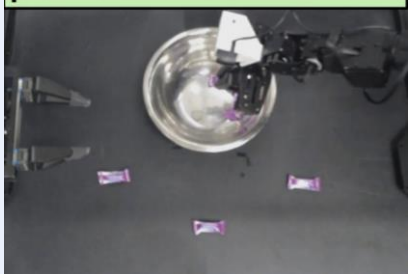
Example: TD-MPC2 on 80-task multi-embodiment control

→ Part 3 introduces a third option: explicit + pre-trained on internet video.

Two priors, two kinds of knowledge

VLM

put candies in bowl



"this is a coffee cup"

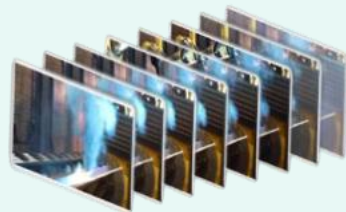
"this is a screwdriver"

"the person is sitting"

Semantics

VS

Video Model



"liquid pours and splashes"

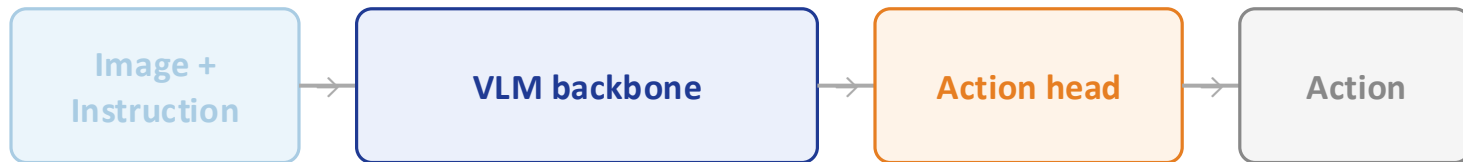
"doors swing open"

"objects fall under gravity"

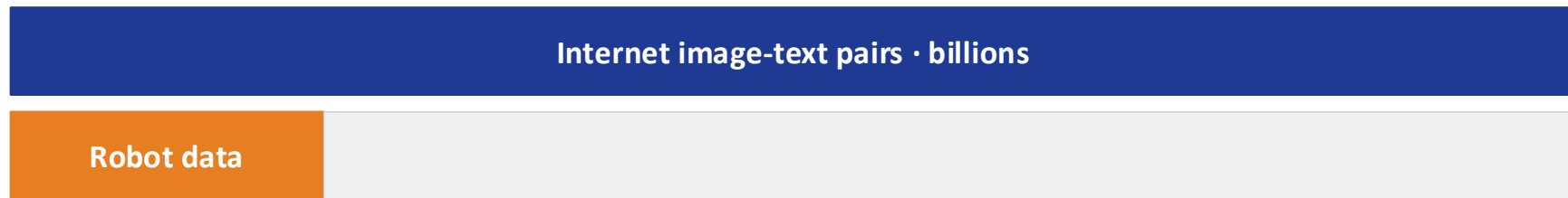
Dynamics

Robotics needs dynamics. Internet video carries dynamics natively.

Today's SOTA VLAs all use a VLM backbone



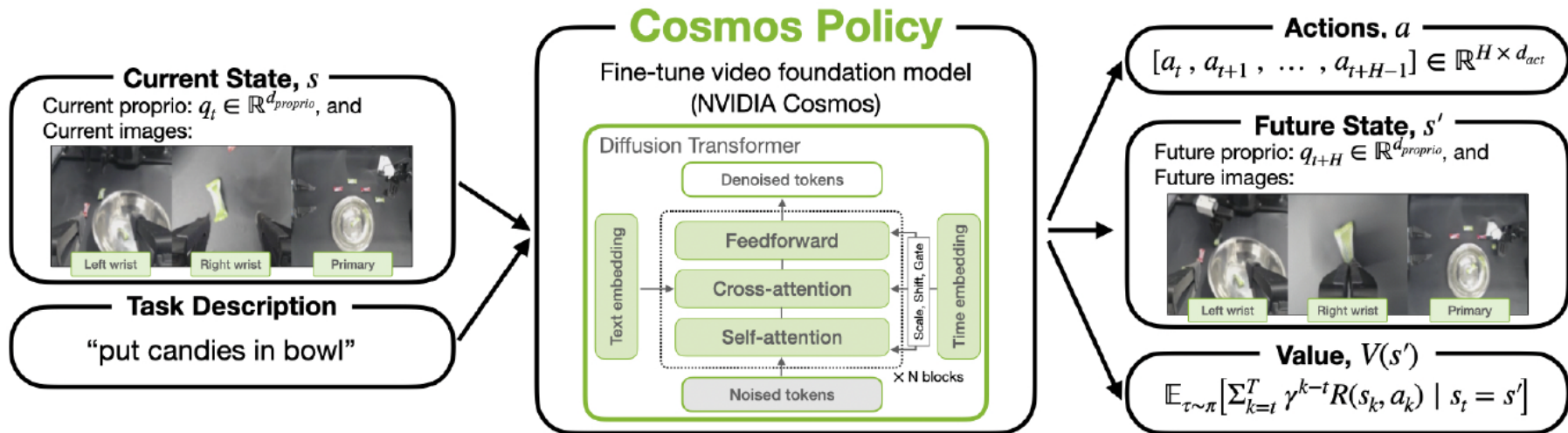
Training data scale:



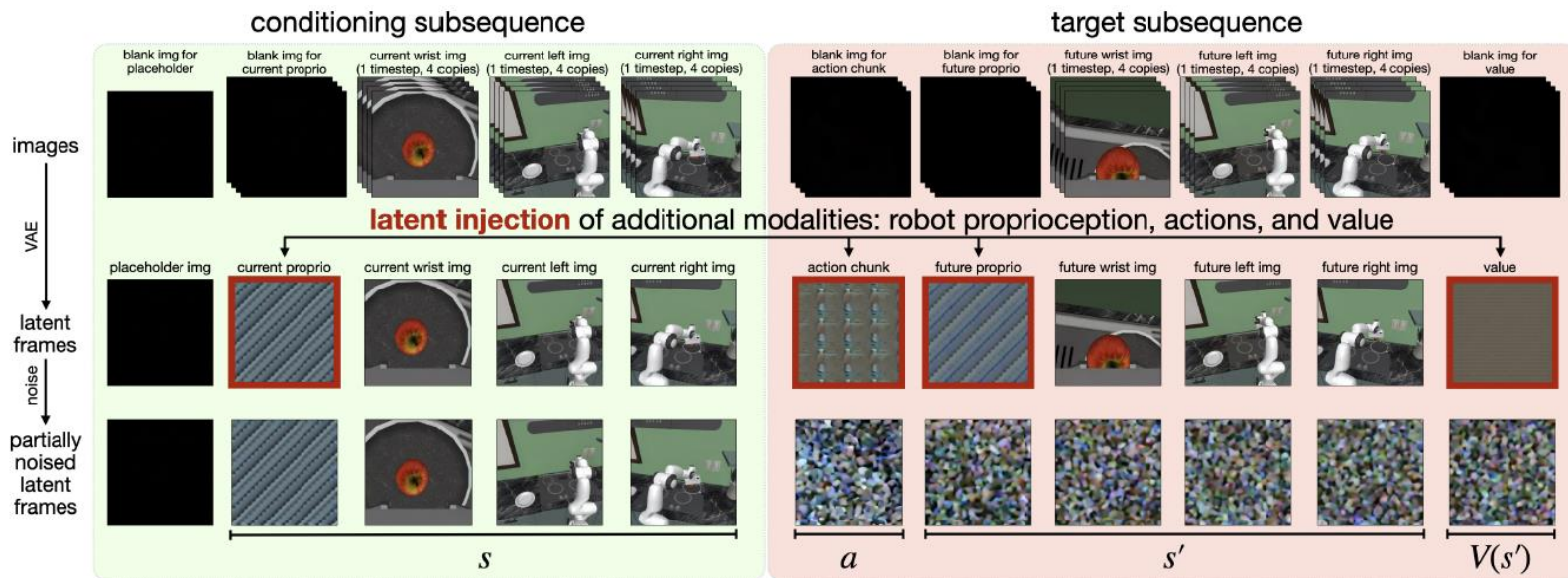
Semantics generalises. Dynamics has to be re-learned from a much smaller dataset.

→ Replace VLM with a video model

Cosmos Policy · video diffusion model as the entire policy stack



Latent frame injection · how the modalities sit in one sequence



Policy mode

→ denoise actions only (parallel)

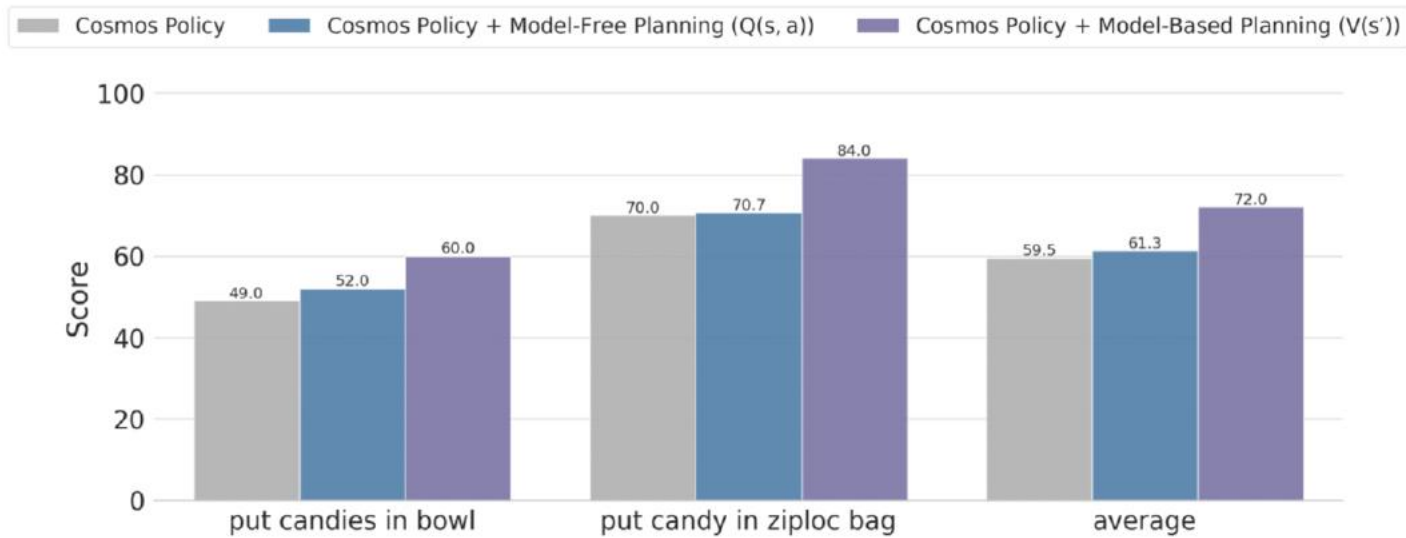
World-model mode

→ autoregressive image rollout

Value mode

→ denoise value latent

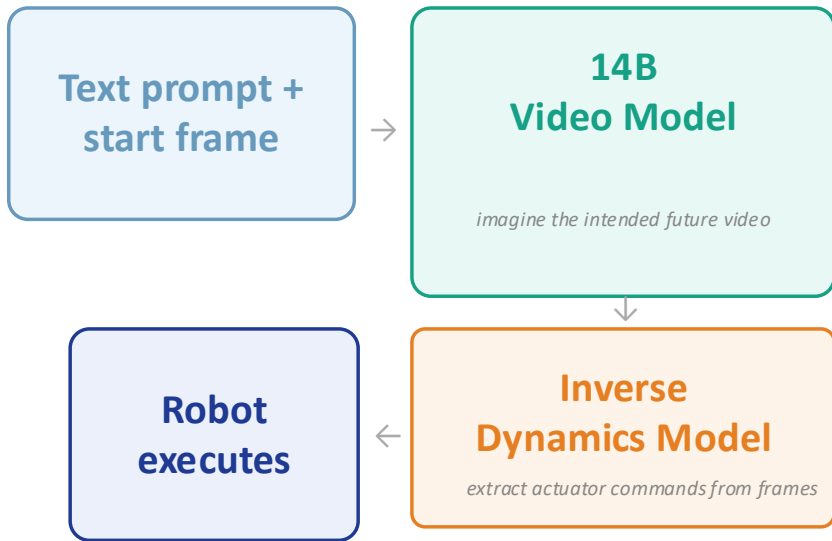
Cosmos Policy · empirical results



Diffusion captures action multimodality natively.

1X NEO · the same idea, shipping today

1X World Model



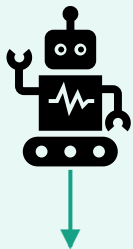
Two differences from Cosmos: (1) actions via external IDM, (2) rejection sampling on kinematic feasibility.

Whoever owns the video prior owns the dynamics.

→ Part 4 · what if the world model is the product?

Flip the picture · the world model as the product

Until now



World Model

World model is the agent's scratchpad.

Part 4



Policy evaluation



Live
teleoperation

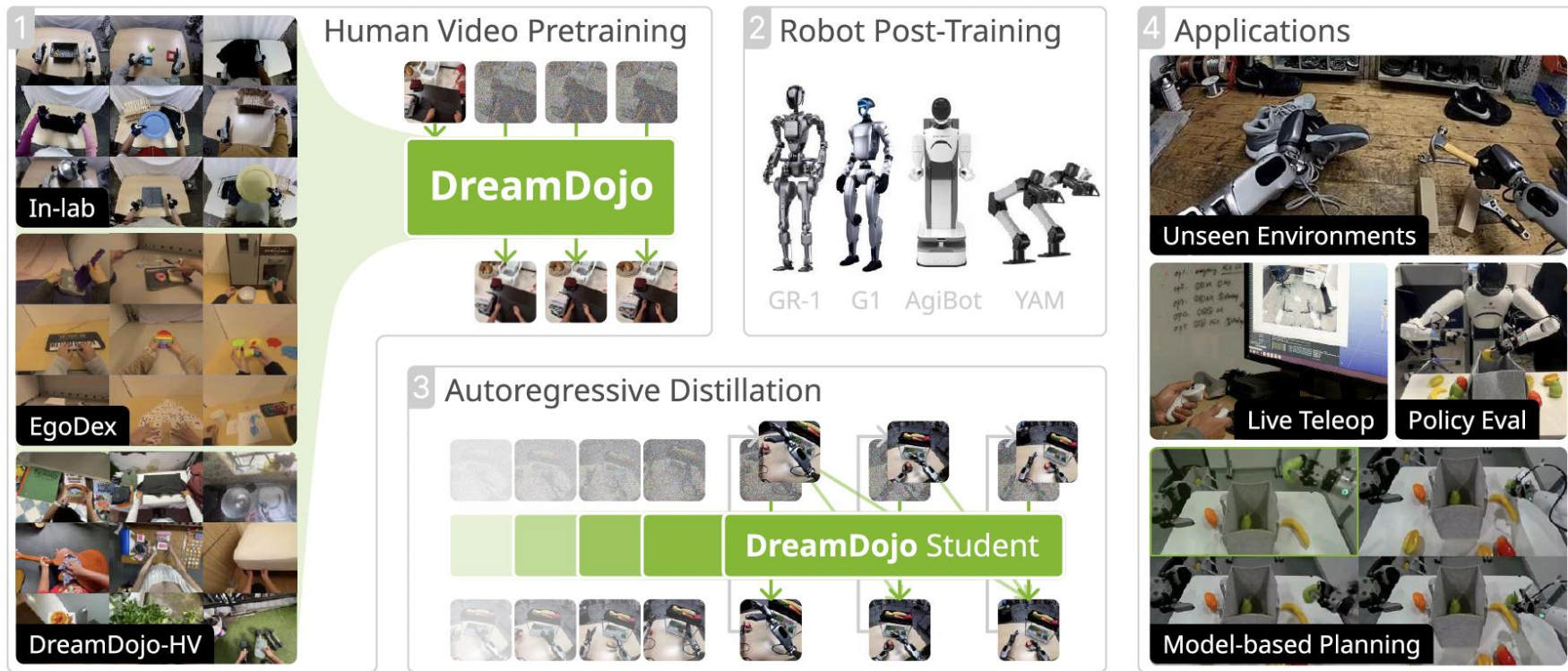


Model-based
planning

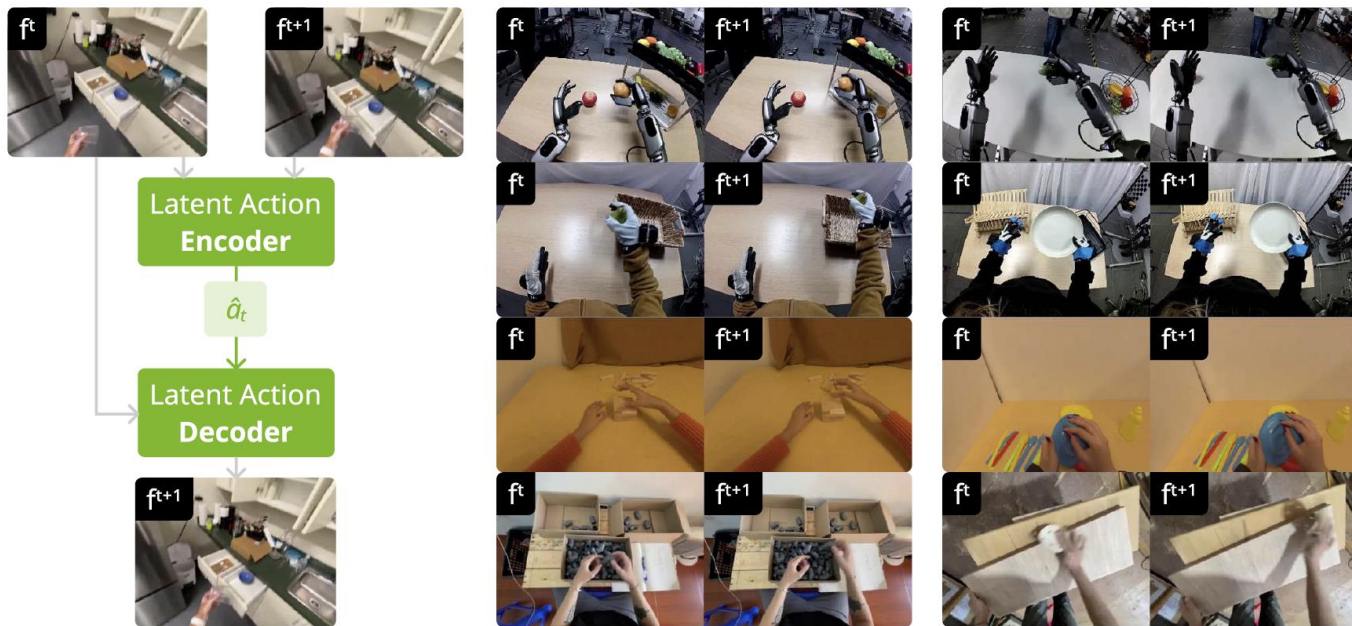
World model is the product. Many agents plug in.

Any policy. Any human pilot. Any researcher. Same world model.

DreamDojo

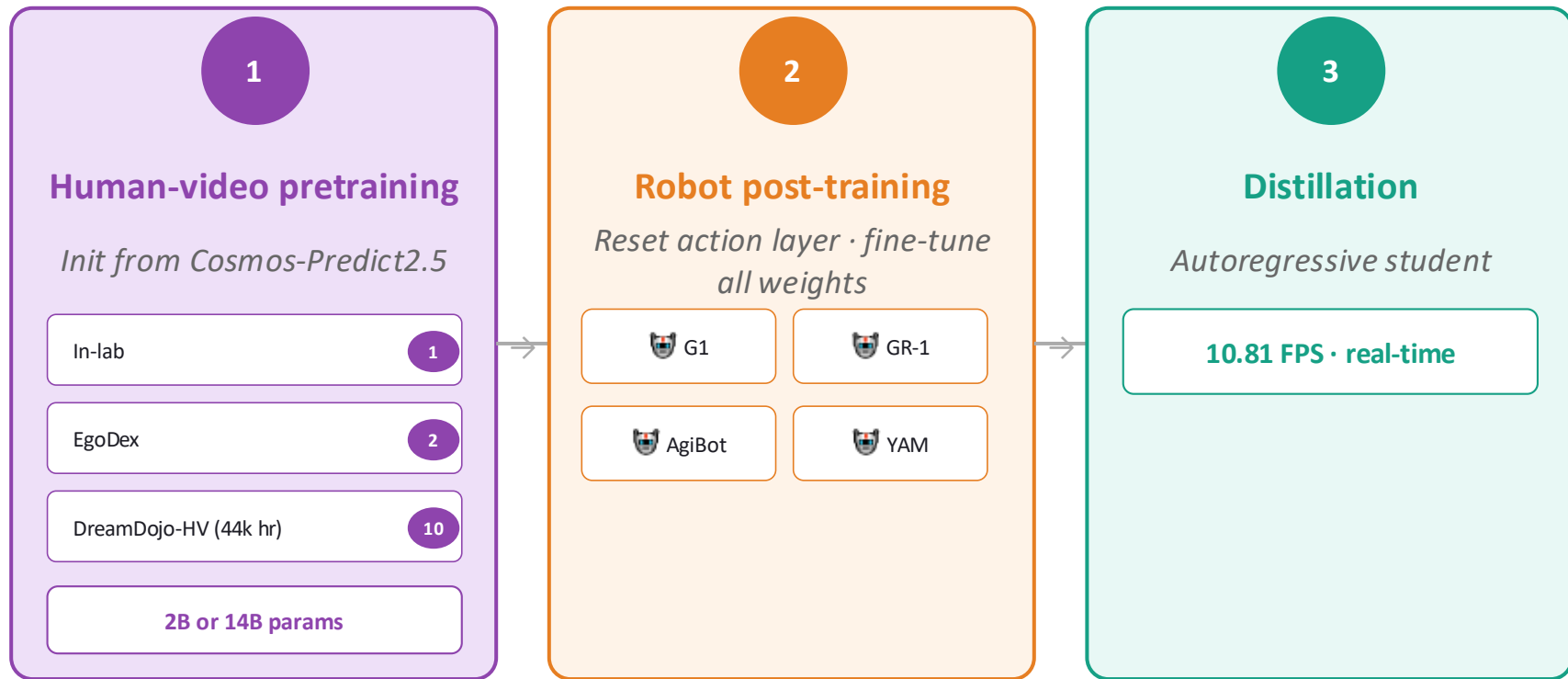


Learning actions from action-free video



Information bottleneck → the latent must contain only what changed between frames → the action.

DreamDojo · three-stage recipe



Same backbone family as Cosmos Policy. Same video diffusion pretraining.

Three applications · the world model as infrastructure



Policy evaluation without a robot

100 OOD scenarios in GPU time. No hardware. No supervisor.



Live teleoperation in synthetic worlds

Human pilot drives a robot through a scene that does not physically exist.

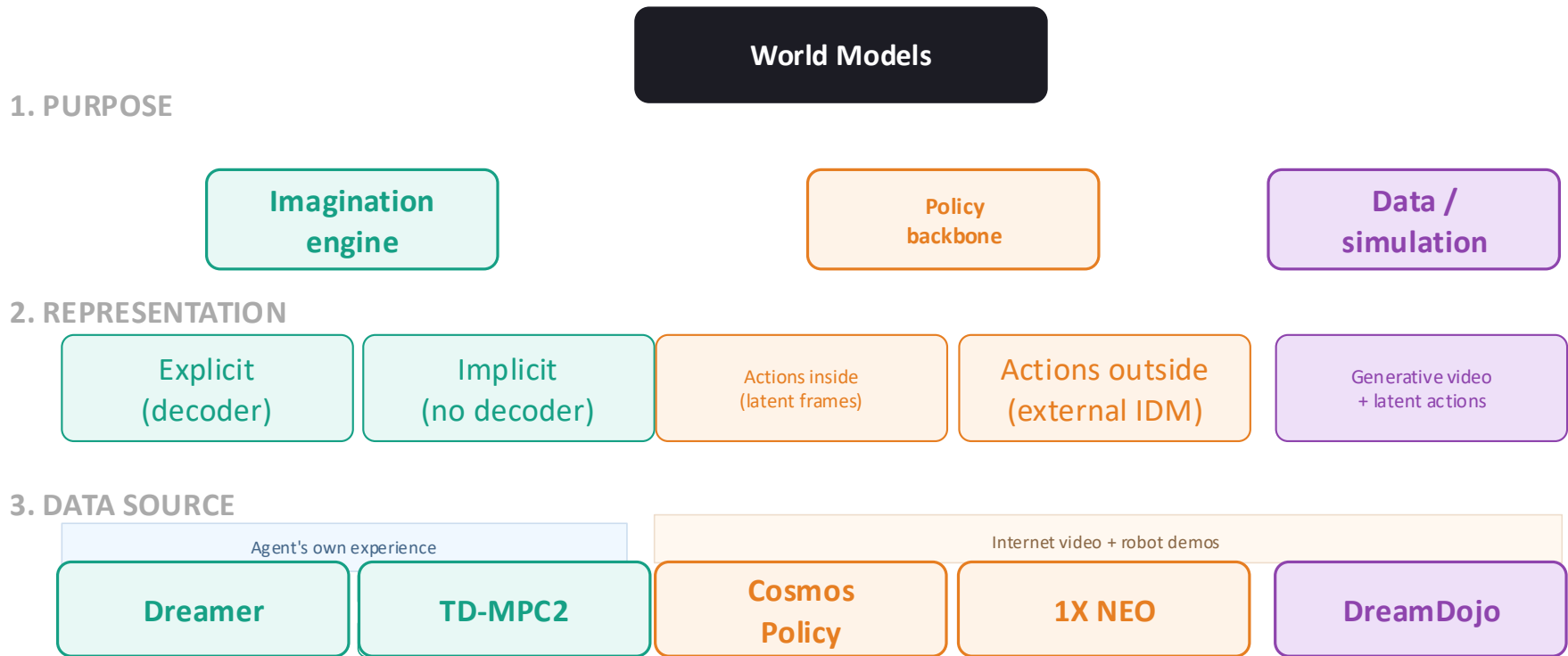


Online model-based planning

Best-of-N at deployment, with an internet-pretrained predictive model.

Connection to Part 1 — SafeDreamer's slogan at internet scale: dangerous imagination, safe execution.

The big picture · one map for the field



Questions?

References



DreamerV3 — Hafner et al., 2023 — [arXiv:2301.04104](#)



Disagreement — Pathak et al., 2019 — [arXiv:1906.04161](#)



TD-MPC2 — Hansen et al., ICLR 2024 — [arXiv:2310.16828](#)



DreamDojo — NVIDIA 2026 — [arXiv:2602.06949](#)



Director — Hafner et al., 2022 — [arXiv:2206.04114](#)



SafeDreamer — Huang et al., ICLR 2024 — [arXiv:2307.07176](#)



Cosmos Policy — Kim et al., NVIDIA 2026 — [arXiv:2601.16163](#)



1X World Model — 1X Technologies, Jan 2026 — [1x.tech/discover/world-model](#)